# AgriRent AI — demo & test guide

Farm-equipment rental for Indian farmers: find a machine for a job, book it, and prove what condition it was in — so neither side can lie about damage.

## Sign in with these two accounts

| Role | Email | Password | What they do on stage |
|---|---|---|---|
| Owner | abdullacyclewala@gmail.com | 12345678 | Lists machines, accepts the booking, records handover, files a claim |
| Renter | menhdi@gmail.com | 12345678 | Searches, books, countersigns, records the return, answers the claim |

**Already set up for you**
Booking **#53** is prepared and ready to walk through: **menhdi@gmail.com** renting **Mahindra 575 DI Tractor** (Abdulla) — status Confirmed, no condition report filed yet, so the handover you shoot on stage is the first real one. Abdulla is in owner mode; Mehndi is in renter mode. Both profiles already have a saved location, so the marketplace shows distances without a permission prompt.

## Links

| What | Where |
|---|---|
| The app | https://agrirent-ai.vercel.app |
| The prepared booking | https://agrirent-ai.vercel.app/booking/53 |
| The new marketplace tab | https://agrirent-ai.vercel.app/marketplace |
| Backend health / AI status | https://agrirent-ai-backend.onrender.com/verification/config |

## Five minutes before you start

- Run the readiness check (read-only, tells you the truth): `SUPABASE_URL=… ANON_KEY=… node scripts/check-demo-ready.mjs`
- Open the app once and log in as **both** accounts in two browser profiles — the free backend sleeps, and the first request after a nap takes 30–60 seconds. Waking it beforehand is the single biggest demo risk.
- Have the machine (or any object) ready to photograph, and allow the camera when the browser asks.

## The demo — 10 minutes, two people, one story

### Part 1 · The app a farmer actually uses (4 min)

| # | Do this | Say this / what to notice |
|---|---|---|
| 1 | Sign in as **menhdi@gmail.com**. Land on the Dashboard. | “Everything starts from the farmer's side: what work do you have?” |
| 2 | Tap **Find Equipment** → type a job in plain language, e.g. “I need to plough 3 acres of wheat in Panvel next week”. | The AI parser fills crop, operation, land size, place and date by itself — the farmer only checks it. |
| 3 | Walk the review step and get results. | A ranking model orders the matches and explains why each one fits. Hard rules (crop, operation, distance) filter first. |
| 4 | Open **Marketplace** (new tab). Filter by type, price, distance; search “Panvel”. | The other half of the product: browse what's nearby without answering a questionnaire. Distance comes from PostGIS and respects each owner's own service radius. |
| 5 | Open a machine → book it for two days. | Booking is a request; the owner accepts. Availability and double-booking are checked server-side. |

| # | Do this | Say this / what to notice |
|---|---|---|
| 6 | Switch to the **Abdulla** window, open the booking, press **Accept**. | Realtime: the renter's page updates without a refresh. (Push notifications can be shown too if enabled.) |

### Part 2 · The part that makes it trustworthy (6 min)

This is the Phase 7 feature: a signed condition report at handover and at return, so “it was already damaged” stops being an argument.

| # | Do this | What to point at |
|---|---|---|
| 7 | As **Abdulla**, open booking #53 and press **Record condition before handover**. | One card, one button — no menu. The deposit is shown as a single number. |
| 8 | Shoot the four guided photos (whole machine ×2, hour meter, tyres/implement). | Each photo is stamped inside the pixels with the report code, date, time and GPS — and fingerprinted twice. Turn the phone: it is four taps. |
| 9 | Note any damage that is already there (chips, one tap), then submit. | This is what protects the renter from being blamed for old damage — the machine gets a file, carried into every future rental. |
| 10 | Read the **“What the app sees in these photos”** panel and sign. | The AI reads the photos and reports damage with a confidence score. It never decides: AI flags, a person decides. The signature locks the record — nobody, including the owner, can change it afterwards. |
| 11 | Switch to **Mehndi**: open the report, press **This is correct**. | One tap. Two separate attestations are now on record — the renter is not signing the owner's word, they are confirming what they saw. |
| 12 | As Mehndi, record the **return** condition the same way. | The return report is compared view-by-view against the handover photos, side by side. |
| 13 | Deliberate mistake (the best 40 seconds of the demo): use the gallery and re-submit the **handover photo** as the return. | The report flags “same image as an earlier photo” and drops to **Not verified**. That is what stops a machine being “inspected” from a desk. |
| 14 | As Abdulla, file a claim of **₹5,000** with no repair estimate, then attach a photographed bill. | The database itself refuses a large claim without an estimate — the rule lives in Postgres, not just in the screen. |
| 15 | As Mehndi, press **Dispute this**. | Nothing is deducted while a claim is disputed: the disputed amount is held, the rest is released. AI never moves money. |
| 16 | Log out, open **/verify/AR-XXXXXXXX** with the report code. | The close: anyone — a bank, an insurer, a court — can check the report is intact, without an account and without seeing anyone's photos. |

## Test it yourself — the short checklist

Each line is: what to do → what you should see. Anything else is a bug; tell me the report code and what happened.

| Area | Do this | Expect |
|---|---|---|
| Accounts | Sign in as both accounts with 12345678 | Both open; Abdulla sees owner tools, Mehndi sees the renter view |
| Marketplace | Open the Marketplace tab; change type/price/distance; search a village name | Cards filter live; distance shown in km; count line updates; “Clear filters” resets |
| Marketplace | Press “Show what's near me” and allow location | Distances refresh and are remembered on your profile |
| Handover | Booking #53 → Record condition before handover → 4 photos → submit | Photos upload with progress; no typing required |

| Area | Do this | Expect |
|---|---|---|
| Capture quality | Take one deliberately dark or blank photo | “Retake suggested” appears — it advises, it never blocks |
| Capture privacy | Deny the location permission on one photo | “No location — the rest still works”; the flow continues |
| AI reading | Look at the AI panel after uploading | Summary + damage items with %; an hour-meter value if the digits are legible |
| AI retry | If it says “AI reading off”, press Re-run the check | The reading appears using the photos already stored — nothing to retake |
| Signature | Sign with a finger, then reload the page | Signature persists; the report is locked and cannot be edited |
| Two parties | Counterparty view: press “This is correct” | One tap records their attestation; the report stops being marked “One-sided” |
| Return | Open the return report | “Before and after, view by view” sliders appear when a handover exists |
| Duplicate trap | Submit the handover photo again as a return photo | Flagged as the same image; verdict drops to Not verified |
| Claim rule | File ₹5,000 with no estimate attached | Refused — the estimate photo is required from ₹2,000 up |
| Deposit | Dispute a claim | ₹0 deducted, disputed amount held; neither party can unilaterally take money |
| Verification | With/AR-XXXXXXXX logged out | “This report is intact”, with entries, signers and fingerprint. A wrong code says “not found” |
| Privacy | As a third account, open the report URL | Sees nothing — RLS keeps evidence to the two parties |
| Languages | Switch to हिंदी / मराठी | Every new screen is translated (Hindi/Marathi copy is machine-drafted — worth a native-speaker read) |

## If something goes wrong on stage

| Symptom | Do this |
|---|---|
| A page spins for 30–60 seconds on the first click | That is the free backend waking up. Say so — it is honest and it is fine. Keep clicking; the next request is instant. |
| “AI reading off — evidence still recorded” | Press **Re-run the check** in that panel. If it still fails, continue: the evidence, signatures and verification all work without any AI — that is by design, and worth saying out loud. |
| The marketplace looks empty | Widen the distance filter (50 → 100 km). An owner's own service radius also limits what appears — that is the rule, not a bug. |
| Camera does not open | Use **Use my gallery**. The flow accepts a gallery photo and notes that it carries no camera metadata — a real, explainable behaviour. |
| A photo is blurry / the retake prompt keeps appearing | Take it again with more light and both hands steady; the prompt is advice. You can also keep it — the report will carry a quality note rather than hide it. |
| Anything else | Note the report code (AR-XXXXXXXX) or booking number, take a screenshot, carry on. Every state is recoverable and nothing is lost. |

## Questions professors usually ask — the honest answers

- **“Can the AI be wrong?”** Yes, and it is built that way on purpose. The AI only flags; every flag is shown with its reason and a person decides. A wrong AI note costs a conversation, not a payment.

- **“What stops someone faking photos?”** No single check would. Five weak signals are combined instead: the exact-byte fingerprint, a perceptual image fingerprint (catches the same photo resized or re-compressed), GPS against the machine's registered location, the camera clock against the server's receipt time, and a watermark burned into the pixels. Faking all five is more work than telling the truth.
- **“Can the record be edited later?”** No. Evidence rows and the audit trail are append-only in the database (no update or delete policy exists), and each entry is chained to the previous one by a SHA-256 hash computed inside Postgres. The public page recomputes that chain — so tampering is detectable, not just forbidden.
- **“Who can see the photos?”** Only the two parties on that booking. The bucket is private, access is enforced by row-level security, and the public verification page shows no photos and no personal data.
- **“Does the AI decide who pays?”** Never. AI cannot set an amount, and while a claim is disputed nothing is deducted. Payment still happens between the two people; the app records the agreement and the policy version it was made under.
- **“What is not finished?”** Payment is still off-platform (deliberate — no payment gateway). There is no persistent offline queue yet: a failed upload retries in the page, but a reload while offline loses that capture. And the Hindi/Marathi copy for this feature is machine-drafted pending a native-speaker review.

**Scope, stated plainly**
Verified condition reports, the dispute and deposit flow, the public verification page and the marketplace are all live on the deployed app. The engineering is tested three ways: 39 logic assertions, 31 backend tests, 43 assertions run against the live database as real signed-in users, plus two live probes of the deployed AI path.

Full detail: docs/PHASE7_MANUAL_TEST_PLAN.md (long-form test script) · docs/PHASE7_VERIFIED_CONDITION_REPORTS.md (design, deviations, limits) · AGRIRENT_AI_MASTER.md §0 (status of every phase).